Assignment 1: Basic Notions of Probability

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Question 1

Let

$$\Omega = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 100\}.$$

Event A:

$$A = \{(x, y) \in \Omega : 9 \leq \sqrt{x^2 + y^2} \leq 10\}.$$

(i) Uniform over area

Total area:

$$100\pi$$

Area of A:

$$100\pi - 81\pi = 19\pi$$

$$P(A) = \frac{19}{100}$$

(ii) Uniform radius

Let $r \sim \text{Uniform}(0, 10)$.

$$P(A) = P(9 \leq r \leq 10) = \frac{1}{10}$$

(iii) Explanation

The first model is uniform over area, while the second is uniform in radius. Since outer regions have larger area, the probabilities differ.

Question 2

Consider a circle of radius R. We find the probability that a random chord is longer than the side of the inscribed equilateral triangle.

The side length of the triangle is

$$\sqrt{3}R.$$

Method 1: Random endpoints

Fix one endpoint and choose the other uniformly on the circumference.

The chord is longer than the triangle side if the central angle exceeds $120°$. The favorable arc is one-third of the circle.

$$P = \frac{1}{3}.$$

Method 2: Random midpoint

The midpoint is chosen uniformly inside the circle.

The chord is long enough if the midpoint lies within a circle of radius $R/2$.

$$P = \frac{\pi(R/2)^2}{\pi R^2} = \frac{1}{4}.$$

Method 3: Random distance from center

A direction is chosen and a point is selected uniformly along the radius.

The chord satisfies the condition if the distance from the center is less than $R/2$.

$$P = \frac{R/2}{R} = \frac{1}{2}.$$

Summary

Method	Probability
Random endpoints	$\frac{1}{3}$
Random midpoint	$\frac{1}{4}$
Random distance	$\frac{1}{2}$

In conclusion, different methods of choosing a random chord lead to different probability distributions. Therefore, the answer depends on the definition of randomness used.

Question 8

Let the sample space for two children be:

$$S = \{BB, BG, GB, GG\}$$

Each outcome is equally likely with probability $\frac{1}{4}$.

Given that at least one child is a girl, we remove BB. So the reduced sample space is:

$$S' = \{BG, GB, GG\}$$

We are required to find:

$$P(\text{both girls} \mid \text{at least one girl})$$

This becomes:

$$P(GG \mid S') = \frac{\text{Number of favorable outcomes}}{\text{Total outcomes in } S'}$$

$$P(GG \mid S') = \frac{1}{3}$$

$$\boxed{\frac{1}{3}}$$

Question 9

1. Probability space

The sample space is

$$\Omega = \{G_1, G_2, D_1, D_2\}, \quad \mathfrak{F} = \mathfrak{P}(\Omega)$$

with probability measure

$$P(G_1) = 0.50, \quad P(G_2) = 0.20, \quad P(D_1) = 0.10, \quad P(D_2) = 0.20$$

We verify that

$$0.50 + 0.20 + 0.10 + 0.20 = 1$$

so this defines a valid probability space.

2. $\sigma(X)$

The random variable X is defined as:

$$X(G_1) = 0, \quad X(G_2) = 0, \quad X(D_1) = 1, \quad X(D_2) = 1$$

So the inverse images are

$$X^{-1}(0) = \{G_1, G_2\}, \quad X^{-1}(1) = \{D_1, D_2\}$$

Hence the σ-algebra generated by X is

$$\sigma(X) = \{\varnothing, \Omega, \{G_1, G_2\}, \{D_1, D_2\}\}$$

This shows that X only distinguishes between good and defective items, but not the machine.

3. $\sigma(Y)$

The random variable Y is defined by:

$$Y(G_1) = 1, \quad Y(G_2) = 2, \quad Y(D_1) = 3, \quad Y(D_2) = 4$$

Since each outcome is assigned a unique value, Y separates all points in Ω. Therefore,

$$\sigma(Y) = \mathfrak{P}(\Omega)$$

This means Y contains full information about every outcome.

4. Inclusion of σ-algebras

Since $\sigma(Y) = \mathfrak{P}(\Omega)$,

$$\sigma(X) \subseteq \sigma(Y)$$

Thus Y is strictly more informative than X, because it distinguishes machine and quality, whereas X only distinguishes quality.

5. Marginal distribution of X

We compute:

$$P(X = 0) = P(G_1) + P(G_2) = 0.70$$
$$P(X = 1) = P(D_1) + P(D_2) = 0.30$$

So the distribution of X is:

$$P(X = 0) = 0.70, \quad P(X = 1) = 0.30$$

6. Marginal distribution of Y

Each value corresponds directly to one outcome:

$$P(Y = 1) = 0.50, \quad P(Y = 2) = 0.20, \quad P(Y = 3) = 0.10, \quad P(Y = 4) = 0.20$$

7. Joint probabilities

Since each pair (X, Y) corresponds to a unique outcome,

$$P(X = 0, Y = 1) = 0.50, \quad P(X = 0, Y = 2) = 0.20$$

$$P(X = 1, Y = 3) = 0.10, \quad P(X = 1, Y = 4) = 0.20$$

All other joint probabilities are zero because no other combinations occur.

8. Conditional probabilities

We compute:

$$P(X = 1 \mid Y = 3) = 1, \quad P(X = 1 \mid Y = 4) = 1$$

because both $Y = 3$ and $Y = 4$ correspond to defective items.
Now,

$$P(Y = 3 \mid X = 1) = \frac{0.10}{0.30} = \frac{1}{3}$$
$$P(Y = 4 \mid X = 1) = \frac{0.20}{0.30} = \frac{2}{3}$$

So even if we know the item is defective, we still do not know which machine produced it.

9. Interpretation

Conditioning on Y removes all uncertainty about X, since Y identifies the exact outcome.

However, conditioning on X still leaves uncertainty about Y, because knowing only whether the item is defective does not tell us the machine used.

10. Information in $\sigma(X)$ and $\sigma(Y)$

The σ-algebra $\sigma(X)$ only contains information about whether the item is good or defective:

$$\{G_1, G_2\} \text{ or } \{D_1, D_2\}$$

On the other hand, $\sigma(Y)$ distinguishes every single outcome:

$$\{G_1\}, \{G_2\}, \{D_1\}, \{D_2\}$$

Hence,

$$\sigma(Y) \text{ is strictly finer than } \sigma(X)$$

so Y contains strictly more information than X.

11. Python visualization

```
import matplotlib.pyplot as plt
import numpy as np

omega = ["G1", "G2", "D1", "D2"]
p = np.array([0.50, 0.20, 0.10, 0.20])

joint = np.array([
    [0.50, 0.20, 0.00, 0.00],
    [0.00, 0.00, 0.10, 0.20]
])

fig, ax = plt.subplots()
im = ax.imshow(joint, cmap="YlOrRd", aspect="auto")

ax.set_xticks(range(4))
ax.set_xticklabels([1, 2, 3, 4])
ax.set_yticks(range(2))
ax.set_yticklabels([0, 1])

ax.set_xlabel("Y")
ax.set_ylabel("X")
ax.set_title("Joint Distribution of (X,Y)")

for i in range(2):
    for j in range(4):
        ax.text(j, i, f"{joint[i,j]:.2f}",
                ha="center", va="center")

plt.colorbar(im)
plt.show()
```

12. Entropy calculations

Using natural logarithms,

$$H(Z) = -\sum P(z) \ln P(z)$$

For X:

$$H(X) = -(0.70 \ln 0.70 + 0.30 \ln 0.30)$$

$$H(X) \approx 0.61086$$

For Y:

$$H(Y) = -(0.50 \ln 0.50 + 0.20 \ln 0.20 + 0.10 \ln 0.10 + 0.20 \ln 0.20)$$

$$H(Y) \approx 1.22061$$

Joint entropy:

$$H(X, Y) = -(0.50 \ln 0.50 + 0.20 \ln 0.20 + 0.10 \ln 0.10 + 0.20 \ln 0.20)$$

$$H(X, Y) \approx 1.22061$$

Since Y determines X:

$$H(X \mid Y) = 0$$

Conditional entropy:

$$H(Y \mid X) = H(X, Y) - H(X)$$

$$H(Y \mid X) \approx 1.22061 - 0.61086 = 0.60975$$

Summary

$$H(X) \approx 0.61086$$

$$H(Y) \approx 1.22061$$

$$H(X, Y) \approx 1.22061$$

$$H(X \mid Y) = 0$$

$$H(Y \mid X) \approx 0.60975$$

Interpretation

$H(X)$ measures uncertainty in quality only.
$H(Y)$ measures uncertainty in both quality and machine.
Since $H(X \mid Y) = 0$, knowing Y removes all uncertainty about X.
Since $H(Y \mid X) > 0$, knowing X still leaves uncertainty about the machine.
Thus, Y contains strictly more information than X.

Question 10

1. Probability space

The probability space is

$$\Omega = \mathbb{R}^2, \qquad \mathfrak{F} = \mathfrak{B}(\mathbb{R}^2)$$

A sample point is

$$\omega = (t, p)$$

where t is the temperature deviation and p is the pressure deviation.

The probability measure is defined by the joint density

$$f_{T,P}(t,p) = \frac{1}{4\pi} \exp\left(-\frac{t^2}{2} - \frac{p^2}{8}\right)$$

So for any Borel set A,

$$P(A) = \iint_A f_{T,P}(t,p)\, dt\, dp$$

Since $T \sim N(0,1)$ and $P \sim N(0,4)$ are independent, this defines a valid probability space.

2. $\sigma(X)$

The random variable is

$$X(t,p) = t$$

So X depends only on temperature.

For any Borel set B,

$$X^{-1}(B) = B \times \mathbb{R}$$

Thus,

$$\sigma(X) = \{B \times \mathbb{R} : B \in \mathfrak{B}(\mathbb{R})\}$$

So $\sigma(X)$ contains all events depending only on temperature.

3. $\sigma(Y)$

The random variable

$$Y(t,p) = (t,p)$$

is the identity map on $\mathbb{R}^2$, hence

$$\sigma(Y) = \mathfrak{B}(\mathbb{R}^2)$$

So Y captures all measurable information in the system.

4. Inclusion of σ-algebras

Since every set of the form $B \times \mathbb{R}$ is Borel,

$$\sigma(X) \subset \sigma(Y)$$

The inclusion is strict because sets like $[-1,1] \times [-2,2]$ are in $\sigma(Y)$ but not in $\sigma(X)$.

Thus, Y contains strictly more information than X.

5. Marginal density of X

$$f_X(x) = \int_{-\infty}^{\infty} f_{T,P}(x,p)\,dp$$

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

So,

$$X \sim N(0,1)$$

6. Marginal density of Y

$$f_Y(t,p) = \frac{1}{4\pi} \exp\left(-\frac{t^2}{2} - \frac{p^2}{8}\right)$$

Thus $Y = (T,P)$ is a bivariate normal with independent components.

7. Required probabilities

$$P(X \leq 0) = \frac{1}{2}$$

$$P(X > 1) = 1 - \Phi(1) \approx 0.1587$$

Since X depends only on T,

$$P(Y \in (-\infty, 0] \times \mathbb{R}) = \frac{1}{2}$$

For the rectangle event:

$$P(Y \in [-1,1] \times [-2,2])$$

Using independence:

$$= P(-1 \leq T \leq 1)\, P(-2 \leq P \leq 2)$$

$$= (2\Phi(1) - 1)^2 \approx 0.466$$

8. Conditional distribution of X given Y

Since $Y = (t, p)$ determines $X = t$ exactly,

$$X \mid Y = (t, p) = \delta_t$$

So the conditional distribution is degenerate.

9. Conditional distribution of Y given X

Given $X = t$, we have

$$Y \mid X = t = (t, P)$$

and

$$P \sim N(0, 4)$$

Thus,

$$f_{Y|X}((u, v) \mid t) = \delta_t(u) \cdot \frac{1}{\sqrt{8\pi}} e^{-v^2/8}$$

10. Interpretation

Conditioning on X gives only temperature information.

Conditioning on Y gives both temperature and pressure.

So Y is strictly more informative than X.

11. Measurability relation

Let

$$\pi_1(t, p) = t$$

Then

$$X = \pi_1 \circ Y$$

So X is measurable with respect to $\sigma(Y)$, hence

$$\sigma(X) \subseteq \sigma(Y)$$

12. Python visualization

```
import numpy as np
import matplotlib.pyplot as plt

# Grid
t = np.linspace(-4, 4, 400)
p = np.linspace(-8, 8, 400)
T, P = np.meshgrid(t, p)

# Joint density
joint = (1/(4*np.pi)) * np.exp(-T**2/2 - P**2/8)
```

```
# Marginal of X
fx = (1/np.sqrt(2*np.pi)) * np.exp(-t**2/2)

# Plot joint density
plt.figure()
plt.contourf(T, P, joint, levels=30, cmap="viridis")
plt.colorbar()
plt.title("Joint density of (T,P)")
plt.xlabel("t")
plt.ylabel("p")
plt.show()

# Marginal of X
plt.figure()
plt.plot(t, fx)
plt.title("Marginal density of X")
plt.xlabel("x")
plt.ylabel("f_X(x)")
plt.show()

# (X) visualization (vertical strip)
plt.figure()
plt.axvspan(-1, 1, alpha=0.3, color="blue")
plt.title("Event in (X): B × R (vertical strip)")
plt.xlabel("t")
plt.ylabel("p")
plt.show()

# (Y) visualization (2D region)
plt.figure()
rect = plt.Rectangle((-1, -2), 2, 4, alpha=0.3, color="green")
plt.gca().add_patch(rect)
plt.title("Event in (Y): full 2D region")
plt.xlabel("t")
plt.ylabel("p")
plt.xlim(-4,4)
plt.ylim(-8,8)
plt.show()
```

13. Differential entropy of X

Since $X \sim N(0,1)$,

$$h(X) = \frac{1}{2}\ln(2\pi e) \approx 1.4189$$

14. Differential entropy of Y

Since independent normals:

$$h(Y) = h(T) + h(P)$$

$$h(T) = \frac{1}{2}\ln(2\pi e), \quad h(P) = \frac{1}{2}\ln(8\pi e)$$

Thus,

$$h(Y) = \ln(2\pi e) + \ln 2 \approx 3.5310$$

15. Conditional entropy $h(Y|X)$

$$h(Y|X) = h(P)$$

$$h(Y|X) = \frac{1}{2}\ln(8\pi e) \approx 2.1121$$

16. Discussion of $h(X|Y)$

Since X is fully determined by Y, the conditional distribution is degenerate.

In continuous systems, this leads to singular behavior, so:

$$h(X|Y) \text{ is not analogous to the discrete case}$$

In discrete case:

$$H(X|Y) = 0$$

In continuous case, entropy behaves differently due to density collapse.

Conclusion

We have:

$$\sigma(X) \subset \sigma(Y), \qquad X = \pi_1 \circ Y$$

So Y is strictly more informative than X, which is confirmed by both measure-theoretic structure and entropy analysis.